
Assignment Overview

Build an AI-powered CSV Importer that can intelligently extract CRM lead information from **any valid CSV format**.

The challenge is **not** parsing CSV files.

The challenge is allowing users to upload CSVs with **different column names, layouts, and structures**, while your system accurately maps and extracts the required CRM fields using AI.

For example, these should all work:

- Facebook Lead Export
- Google Ads Export
- Excel sheets
- Real Estate CRM exports
- Sales reports
- Marketing agency CSVs
- Manually created spreadsheets

Your application should identify the appropriate fields and convert them into GrowEasy CRM format.

Functional Requirements

The project consists of both a Frontend and Backend.

Frontend Requirements

Create a responsive web application.

Step 1 — Upload CSV

Allow users to upload a valid CSV file.

Examples:

- Drag & Drop upload
 - File Picker
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Step 2 — Preview

After upload:

- Parse the CSV
- Show a preview of uploaded rows
- Display data inside a beautiful responsive table
- Table should support:
 - Horizontal scrolling
 - Vertical scrolling
 - Sticky headers (preferred)
 - Responsive design

No AI processing should happen yet.

Step 3 — Confirm Import

Provide a **Confirm** button.

Only after confirmation should the frontend call the backend API.

Step 4 — Display Parsed Result

Backend returns AI-extracted CRM records.

Display the parsed result in another responsive table.

Show:

- Successfully parsed records
 - Skipped records (if any)
 - Total imported
 - Total skipped
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Backend Requirements

Create APIs that:

1. Accept CSV Upload

Accept any valid CSV file.

Do **not** assume column names are fixed.

2. Parse CSV

Convert the CSV into records.

3. AI Extraction

Send records to an AI model in batches.

The AI should intelligently map available fields into GrowEasy CRM format.

You may use OpenAI, Gemini, Claude, or any equivalent LLM.

4. Return Structured JSON

Return extracted CRM records in JSON format.

CRM Fields

Your AI should extract as many of the following fields as possible.

Field	Description
created_at	Lead creation date
name	Lead name
email	Primary email
country_code	Country code

mobile_without_country_code	Mobile number
company	Company name
city	City
state	State
country	Country
lead_owner	Lead owner
crm_status	Lead status
crm_note	Notes/remarks
data_source	Source
possession_time	Property possession time
description	Additional description

Sample CRM Records

created_at,name,email,country_code,mobile_without_country_code,company,city,state,country,lead_owner,crm_status,crm_note,data_source,possession_time,description

2026-05-13 14:20:48,John

Doe,john.doe@example.com,+91,9876543210,GrowEasy,Mumbai,Maharashtra,India,test@gmail.com,GOOD_LEAD_FOLLOW_UP,Client is asking to reschedule demo,,,

2026-05-13 14:25:30,Sarah Johnson,sarah.johnson@example.com,+91,9876543211,Tech Solutions,Bangalore,Karnataka,India,test@gmail.com,DID_NOT_CONNECT,"Person was busy, will try again next week",,,

2026-05-13 14:30:15,Rajesh Patel,rajesh.patel@example.com,+91,9876543212,Startup Inc,Delhi,Delhi,India,test@gmail.com,BAD_LEAD,Not interested in our services,,,

2026-05-13 14:35:22,Priya Singh,priya.singh@example.com,+91,9876543213,Enterprise Corp,Pune,Maharashtra,India,test@gmail.com,SALE_DONE,"Deal closed, onboarding in progress",,,

AI Instructions

Your AI should follow these rules while extracting records.

1. Allowed CRM Status Values

Only use one of:

- GOOD_LEAD_FOLLOW_UP
 - DID_NOT_CONNECT
 - BAD_LEAD
 - SALE_DONE
-

2. Allowed Data Source Values

Only use one of:

- leads_on_demand
- meridian_tower
- eden_park
- varah_swamy
- sarjapur_plots

If none match confidently, leave it blank.

3. Date Format

`created_at` must be convertible using JavaScript:

```
new Date(created_at)
```

4. CRM Notes

Use `crm_note` for:

- Remarks
- Follow-up notes
- Additional comments
- Extra phone numbers
- Extra email addresses
- Any useful information that doesn't fit another field

5. Multiple Emails or Mobile Numbers

If multiple email addresses exist:

- Use the first email.
- Append remaining emails into `crm_note`.

If multiple mobile numbers exist:

- Use the first mobile.
 - Append remaining numbers into `crm_note`.
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6. CSV Compatibility

Each record must remain a single CSV row.

Avoid introducing unintended line breaks.

If line breaks are necessary, escape them appropriately (for example, `\n`) so the CSV remains valid.

7. Skip Invalid Records

If a record contains **neither**:

- email

nor

- mobile number

then skip that record.

Evaluation Criteria

Candidates will primarily be evaluated on the following:

AI Prompt Engineering

- Ability to extract fields accurately
 - Intelligent field mapping
 - Handling messy datasets
 - Handling ambiguous columns
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Backend Quality

- API design
 - Clean architecture
 - Error handling
 - Batch processing
 - Maintainable code
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Frontend Quality

- Modern UI
 - Responsive layout
 - Clean UX
 - CSV preview experience
 - Loading states
 - Error handling
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Code Quality

- Readability
 - Type safety
 - Folder structure
 - Reusability
 - Best practices
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Overall Engineering

- Performance
 - Edge case handling
 - Production readiness
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Bonus Points

Additional credit will be given for implementing:

- Drag & Drop upload
 - Progress indicators during AI processing
 - Streaming or incremental parsing
 - Retry mechanism for failed AI batches
 - Virtualized table for large CSVs
 - Dark mode
 - Unit tests
 - Docker setup
 - Deployment using Vercel, Railway, Render, or similar platforms
 - Well-written README with setup instructions
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Tech Stack

Frontend

- Next.js

Backend

- Node.js
- Express

AI

- OpenAI
- Gemini
- Claude
- Any equivalent LLM

Database (Optional)

Use any database if required, or keep the project stateless.

Final Submission Checklist

Before submitting, ensure you have included:

- Publicly hosted application
- Public GitHub repository

- README containing setup instructions
- Position applied for (Intern / Full-Time)

Email everything to:

varun@groweasy.ai

We look forward to reviewing your submission. Good luck!

Screenshots for Reference



